Cyber Security

Curriculum

Axia Africa

Accredited by the American Council of Training and Development

Overview

A 16-week cybersecurity curriculum to provide participants with essential knowledge and technical skills to build a cybersecurity career. The course covers networking, threat analysis, penetration testing, incident response, digital forensics, cryptography, cloud security, governance, risk, and compliance (GRC), security operations, and AI in cybersecurity.

Through interactive lectures, hands-on labs, case studies, and practical projects, learners will develop a solid foundation in cybersecurity principles and industry best practices. The program also emphasizes end-user awareness and data protection to

Expected Outcomes

By the end of this course, participants will be able to:

- Understand cybersecurity fundamentals and the evolving threat landscape.
- Develop hands-on skills in networking, Linux, and security tools.
- Perform vulnerability assessments and penetration testing.
- Implement effective identity and access management (IAM) controls.
- Analyze cyber incidents and conduct digital forensic investigations.
- Understand cryptographic techniques and data protection laws.
- Apply governance, risk, and compliance (GRC) principles to security frameworks.
- Strengthen end-user security awareness to reduce human-targeted cyber risks.
- Understand SOC operations, SIEM tools, and monitoring techniques.

Course Outline

Week	Topics	Learning Objectives
1	**Introduction to Cybersecurity**	▪ Overview of cybersecurity concepts and importance ▪ Understand security principles with a focus on the CIA triad ▪ Explore various domains within cybersecurity ▪ Understand Legal and ethical considerations
2	**Cyber Threat Landscape & Security Fundamentals**	▪ Understand current threat landscape and identify threat actors ▪ Understand phishing, social engineering, and email security ▪ Implement security controls and measures ▪ Discuss the role of security awareness and policies
3	**Operating Systems**	▪ Identify the different types of OS ▪ Explore components and architecture (kernel, user interface, and file systems) ▪ Understand Virtualization technologies ▪ Use Linux commands, bash scripting, and network configurations for basic security tasks
4	**Networking Fundamentals**	▪ Define key networking concepts and protocols ▪ Implement network security protocols ▪ Network mapping and traffic analysis ▪ Configure firewalls, VPNs, IDS/IPS
5	**Cryptography**	▪ Learn to implement cryptographic algorithm ▪ Understand common cryptographic protocols and secure communication channels
6	**Identity & Access Management (IAM)**	▪ Describe core components of IAM ▪ Understand the role of authentication methods ▪ Explore access control models ▪ Implement IAM configurations
7	**Cyber Threat Intelligence**	▪ Explore threat intelligence sources and types ▪ Understand adversary tactics, techniques, and procedures ▪ Explore tools and platforms for threat hunting ▪ Gather and analyze threat intelligence on APTs
8	**Threats, Attacks & Vulnerabilities**	▪ Understand the relationship between threats, ▪ vulnerabilities, and attacks in the attack lifecycle ▪ Identify threats and common attack types ▪ Recognize IOCs and conduct malware analysis

9	**Security Operations Center (SOC) & Monitoring**	▪ Overview of SOC functions and monitoring ▪ Configure SIEM, conduct log and traffic analysis, and perform incident triage ▪ Explore tools for log and traffic analysis
10	**Vulnerability Assessment & Penetration Testing (VAPT)**	▪ Learn vulnerability assessment techniques and penetration test methodologies ▪ Understand the purpose and ethical considerations involved in conducting penetration test ▪ Explore common vulnerabilities and exploits ▪ Identify tools for conducting VAPT
11	**Governance, Risk & Compliance (GRC)**	▪ Understand regulatory compliance requirements (GDPR, HIPAA, PCI-DSS) ▪ Develop security policies and standards ▪ Identify risk management frameworks and processes (NIST RMF, ISO 27001)
12	**Threat detection & Incident Response**	▪ Define and differentiate threat detection methods ▪ Understand incident response phases, create an incident response plan, and analyze past incidents. ▪ Develop incident response playbooks
13	**Digital Forensics & Investigation**	▪ Learn forensic techniques, data acquisition, and evidence handling ▪ Outline the process for collecting and imaging digital evidence ▪ Use forensic tools to perform file system analysis and memory forensics
14	**Cloud Security Fundamentals**	▪ Explore cloud security principles and best practices ▪ Identify common threats in cloud environments ▪ Differentiate cloud service models and division of security responsibilities
15	**AI in Cybersecurity**	▪ Explore AI applications in cybersecurity (threat ▪ detection, automation). ▪ Identify AI algorithms commonly used in threat detection ▪ Integrating AI with traditional cybersecurity tools ▪ Ethical challenges and biases in AI systems
16	**Career Readiness & Capstone Project Preparation**	